1:

Solution:

Proof. We must show that A is open in X. Since $\forall\, x \in A\, \exists U$ open such that $x \in U$ and $U \subset A$ we know that

$$\bigcup_{i \in I} U_i = A.$$

Since each $U_i \subset A \subset X, U_i \subset X$ and $A \subset \bigcup U_i$ since every $x \in A$ is also in some U_i and thus the union. So A must be open in X since the topology is closed under arbitrary union. □

3:

Solution:

Proof. (a) We must show that $\mathcal{T}_C$ is a topology. First by definition of $\mathcal{T}_C$ both $X, \varnothing \in \mathcal{T}_C$. Now, if $\{U_i\}_{i \in I} \in \mathcal{T}_C$ to show that $\bigcup U_i \in \mathcal{T}_C$ we see

$$X - \bigcup U_i = \bigcap (X - U_i)$$

which must be countable since each $X - U_i$ is either all of X or countable (by definition of $\mathcal{T}_C$) and their intersection must also be countable as a subset of countable sets. If $U, V \in \mathcal{T}_C$ are nonempty then $X - (U \cap V) \in \mathcal{T}_C$ since $U \cap V$ is countable (by being a subset of countable sets) and so $X - (U \cap V)$ must also be countable. Thus $\mathcal{T}_C$ is a topology.

(b) No, $\mathcal{T}_\infty$ is not a topology on X because the intersection of infinite sets is not always infinite. Take $X = \mathbb{Z}$ and let $U = \mathbb{Z}^+ \cup \{0\}$ and $V = \mathbb{Z}^- \cup \{0\}$. It is clear that $U, V \in \mathcal{T}_\infty$ but $U \cap V = \{0\} \notin \mathcal{T}_\infty$ and $\mathcal{T}_\infty$ is not closed under finite intersections and thus cannot be a topology.

□

5:

Solution:

Proof. Let $\mathcal{T}_\mathcal{A}$ be the topology generated by $\mathcal{A}$ and let $\mathcal{T}'$ be the intersection of all topologies $\mathcal{T}$ of X that contain $\mathcal{A}$. It is clear by definition that $\mathcal{T}' \subseteq \mathcal{T}_\mathcal{A}$ since $\mathcal{T}_\mathcal{A}$ is apart of the collection of sets being intersected.Now, consider some $U \in \mathcal{T}_\mathcal{A}$. Since $\mathcal{A}$ is a basis, $U = \bigcup A_i$ for A_i basis elements. Now let $\mathcal{T}$ be any topology on X containing $\mathcal{A}$ Since $A \subseteq \mathcal{T}, \mathcal{A}_i \subseteq \mathcal{T}$ and $\mathcal{T}$ is closed under arbitrary union so $U \in \mathcal{T}$ thus $\mathcal{T}_\mathcal{A} \subseteq \mathcal{T}'$ and $\mathcal{T}_\mathcal{A} = \mathcal{T}'$.

Now, let $\mathcal{A}$ be a subbasis and $\mathcal{T}_\mathcal{A}$ be the topology generated by $\mathcal{A}$. As before it is clear that $\mathcal{T}' \subset \mathcal{T}_\mathcal{A}$ since $\mathcal{T}_\mathcal{A}$ belongs to the family of sets being intersected. We define $\mathcal{B}$ to be the collection of all finite intersection of elements of $\mathcal{A}$. Then $\mathcal{T}_\mathcal{A} = \mathcal{T}_\mathcal{B}$ where $\mathcal{T}_\mathcal{B}$ is the topology generated by $\mathcal{B}$. It follows that $\mathcal{B} \subseteq \mathcal{T}$ since $B = \bigcap A_i$ and each $A_i \subset A \in \mathcal{T}$. Applying the idea proven above, $\mathcal{T}_\mathcal{B} = \mathcal{T}_\mathcal{A} = \mathcal{T}'$ since $\mathcal{T}_\mathcal{B}$ is generated by a basis. □

7:

Solution: We know that $\mathcal{T}_1 \subseteq \mathcal{T}_2$ since any basis element $(a, b) \in \mathcal{T}_1$ is automatically a basis element in $\mathcal{T}_2$ by definition. $\mathcal{T}_1 \subseteq \mathcal{T}_4$ as well since for $(a, b) \in \mathcal{T}_1$ and $x \in (a, b)$, there exists $x \in (a, x] \subset (a, b)$ with $(a, x] \in \mathcal{T}_4$. $\mathcal{T}_3 \subseteq \mathcal{T}_1$ since for any $U \in \mathcal{T}_3$ finite, there exists some ε such that $(y-\varepsilon, y+\varepsilon) \subseteq \mathbb{R} \setminus F = U$ for

$F = \mathbb{R} \setminus U$ where $(y-\varepsilon, y+\varepsilon) \cap F = \varnothing$. It is clear that $\mathcal{T}_5 \subseteq \mathcal{T}_1$ since for $x \in (a,b) \in \mathcal{T}_1$, $x \in (-\infty, b) \in \mathcal{T}_5$ and $(a,b) \subseteq (-\infty, a)$. It follows that $\mathcal{T}_5 \subseteq \mathcal{T}_2$ and $\mathcal{T}_5 \subseteq \mathcal{T}_4$ from transitivity of subsets and $\mathcal{T}_3 \nsubseteq \mathcal{T}_5$ and $\mathcal{T}_5 \nsubseteq \mathcal{T}_3$ so $\mathcal{T}_3$ and $\mathcal{T}_5$ are not comparable since the complement of $(-\infty, a)$ is always infinite and the union of infinite sets will always be infinite. $\mathcal{T}_2 \subseteq \mathcal{T}_4$ since for $x \in (a,b) \in \mathcal{T}_2, x \in (x-\varepsilon, x+\varepsilon] \subseteq (a,b)$ for epsilon small so $(a,b) \in \mathcal{T}_4$. For $x \in (a,b) \setminus K$ find $\max\{K \cap (0,x)\} < c < x$ so that $x \in (c,x] \subseteq (a,b) \setminus K$. Now, applying associativity of subsets we can see $\mathcal{T}_3 \subseteq \mathcal{T}_2$ and $\mathcal{T}_3 \subseteq \mathcal{T}_4$.

8:

Solution:

(a) We must show $\mathcal{C}$ is a collection of open sets such that for each $U \subset \mathbb{R}$ each $x \in U \, \exists C \in \mathcal{C}$ such that $x \in C \subset U$. Consider some $U \subset \mathbb{R}$ open and $x \in U$. Then take $C = (x - \varepsilon, x + \varepsilon)$ for ε small and $x \pm \varepsilon \in \mathbb{Q}$ so $C \in \mathcal{C}$ and then $C \subset U$. Thus, by Lemma 13.2 $\mathcal{C}$ is a basis for X. By Lemma 13.1 $\mathcal{T}$ is the union of all elements of $\mathcal{B}$ which is exactly the standard topology.

(b) It suffices to show that $\mathcal{T}_{\mathcal{C}}$ (the topology generated by $\mathcal{C}$) is strictly coarser than $\mathcal{T}_\ell$, i.e., $\mathcal{T}_{\mathcal{C}} \subset \mathcal{T}_\ell$. Consider some basis element $[a,b)$ of $\mathcal{T}_{\mathcal{C}}$ and $x \in [a,b) \in \mathcal{T}_\ell$ since $[a,b)$ is also a basis element of $\mathcal{T}_\ell$. It follows that $\mathcal{T}_{\mathcal{C}} \subset \mathcal{T}_\ell$. We must now show $\mathcal{T}_\ell \nsubseteq \mathcal{T}_{\mathcal{C}}$. Consider $[c,d) \in \mathcal{T}_\ell$ a basis element with $c \in \mathbb{R} \setminus \mathbb{Q}$ and let $x = c \in [c,d)$. Then there is no open set in $\mathcal{T}_{\mathcal{C}}$ such that x is contained in it and it is a subset of $[c,d)$ since all endpoing of $\mathcal{C}$-basis elements are rational. Thus $\mathcal{T}_\ell \nsubseteq \mathcal{T}_{\mathcal{C}}$ and $\mathcal{T}_\ell$ is strictly finer than $\mathcal{T}_{\mathcal{C}}$.